

Voice-Based Health Classification: Capstone Project Report

1. Introduction

The **Voice-Based Health Classification Project** aimed to develop a predictive model capable of identifying whether a patient's voice is classified as *Healthy* or *Unhealthy* based on extracted acoustic features. The dataset used contained preprocessed audio-derived features such as Mel-Frequency Cepstral Coefficients (MFCCs), Spectral Centroid, Root Mean Square Energy (RMSE), and Zero Crossing Rate (ZCR), among others.

Dataset Overview: - **Source File:** `VowelA_High_latest.csv` - **Features:** 26 total, including 20 MFCCs, RMSE, Spectral Centroid, Spectral Bandwidth, Rolloff, and ZCR. - **Metadata:** Gender (G: 'm' or 'w'), Age (A) - **Target Variable:** `label` (Healthy or Unhealthy)

Objective: 1. Explore and clean the dataset through Exploratory Data Analysis (EDA). 2. Visualize key insights and relationships in Power BI. 3. Build and evaluate multiple machine learning models for classification. 4. Present findings and recommendations through an interactive dashboard and written report.

2. EDA Findings

Exploratory Data Analysis (EDA) was conducted using Python libraries (Pandas, Matplotlib, and Seaborn). The analysis focused on understanding feature distributions, correlations, and class imbalances.

Key Insights: - The dataset was balanced across health categories, ensuring fair model training. - Gender distribution was approximately even, though slightly more male participants were observed. - **Age distribution:** Most participants were between 25–55 years. - **Feature behavior:** - Healthy voices showed lower RMSE (energy irregularity) and ZCR values, indicating smoother, stable vocal signals. - Unhealthy voices exhibited higher spectral centroid and entropy, suggesting greater frequency variability. - **Correlations:** MFCC features were moderately correlated with spectral and energy-based attributes, confirming their shared role in representing timbral characteristics.

Pairplots and heatmaps revealed distinct acoustic separations between healthy and unhealthy groups, confirming that vocal degradation manifests in measurable spectral patterns.

3. Data Cleaning

Comprehensive cleaning was performed to ensure model reliability:

- **Missing Values:** None detected; dataset was complete.
- **Outliers:** Mild outliers observed in energy-related features; retained for model generalization.
- **Encoding:**
 - Gender encoded as binary (m = 1, w = 0)
 - Label encoded as (Healthy = 0, Unhealthy = 1)

- **Scaling:** StandardScaler applied to all numeric features for model normalization.
- **Train-Test Split:** 80/20 ratio used to ensure sufficient test coverage.

The final cleaned dataset was stored as `Cleaned_VowelA_High_latest.csv`.

4. Power BI Dashboard

An interactive Power BI dashboard was created to visualize and communicate insights.

Dashboard Pages:

1. Overview Page:

2. Demographic KPIs (sample count, average age)
3. Health status distribution (Healthy vs. Unhealthy)
4. Gender and age-based bar and histogram charts.

5. Acoustic Feature Analysis Page:

6. Comparative visuals showing mean RMSE, Spectral Centroid, and ZCR by health status.
7. Dynamic field parameter allowing feature selection.
8. Scatter plots illustrating relationships between spectral and energy-based variables.

9. Voice Health Summary Page:

10. Narrative cards summarizing key insights (e.g., "Unhealthy voices show higher RMSE and ZCR.")
11. Interactive chart to explore feature differences across age and gender.
12. Text summary box highlighting the strongest differentiators.

13. Model Performance Page:

14. Model selector (Logistic Regression, Decision Tree, Random Forest, XGBoost).
15. KPI cards showing accuracy, precision, recall, F1-score, and AUC.
16. Confusion matrix and ROC curve visuals linked to the selected model.
17. Feature importance chart for tree-based models.

The dashboard emphasizes clarity, interactivity, and narrative storytelling, ensuring technical and non-technical users can interpret results.

5. Predictive Model Development

Machine learning models were implemented and evaluated using Scikit-learn and XGBoost.

Algorithms Tested: - Logistic Regression - Decision Tree Classifier - Random Forest Classifier - XGBoost Classifier

Evaluation Metrics: Accuracy, Precision, Recall, F1-Score, and ROC-AUC were used for model assessment.

Model	Results:	Model	Accuracy	Precision	Recall	F1	AUC
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Logistic Regression	0.87	0.85	0.88	0.86	0.91	0.91	0.91
Decision Tree	0.89	0.88	0.90	0.89	0.92	0.93	0.95
Random Forest	0.91	0.89	0.93	0.91	0.95	0.95	0.95
XGBoost	0.92	0.90	0.94	0.92	0.96	0.96	0.96

Findings: - Ensemble models (Random Forest, XGBoost) outperformed baseline models. - XGBoost achieved the best overall performance (AUC = 0.96), confirming its ability to capture nonlinear patterns. - ROC curves showed clear separation between classes for top models.

Feature Importance: Energy-based (RMSE), spectral (Centroid), and noise-based (ZCR) features were consistently the most influential predictors of vocal health.

6. Conclusion & Future Work

This capstone demonstrated that **voice recordings contain measurable acoustic patterns that can predict vocal health conditions** with high accuracy. Through EDA, Power BI visualization, and machine learning modeling, this project successfully built an interpretable, end-to-end solution.

Key Takeaways: - Unhealthy voices exhibit higher variability in spectral and energy-based features. - XGBoost provided the most accurate classification performance (92% accuracy, 0.96 AUC). - Interactive dashboard effectively visualizes model results and feature behaviors.

Future Improvements: - Incorporate additional vocal samples for broader generalization. - Explore deep learning approaches (e.g., CNNs for raw spectrograms). - Implement explainability tools (e.g., SHAP or LIME) for feature-level interpretation.

Deliverables: - Python Notebooks: `Voice-Based Health EDA.ipynb`, `Predictive_Model.ipynb` -
Cleaned Dataset: `Cleaned_VowelA_High_latest.csv` - Dashboard: `Voice-Based Health.pbix` -
Report: `Voice-Based Health Classification.docx / .pdf`

Author: Christopher Mauro
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